Camera/LiDAR Sensor Fusion-based Autonomous Navigation

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***Abstract*— This research presents a novel approach for autonomous navigation of Unmanned Ground Vehicles (UGV) using a camera and LiDAR sensor fusion system. The proposed method is designed to achieve a high rate of obstacle detection, distance estimation, and obstacle avoidance. In order to thoroughly study the form of things and decrease the problem of object occlusion, which frequently happens in camera-based object recognition, the 3D point cloud received from the LiDAR depth sensors is used. The proposed camera and LiDAR sensor fusion design balance the benefits and drawbacks of the two sensors to produce a detection system that is more reliable than others. The UGV's autonomous navigation system is then provided with the region proposal to re-plan its route and navigate appropriately. The experiments were conducted on a UGV system with high obstacle avoidance and fully autonomous navigation capabilities. The outcomes demonstrate that the suggested technique can successfully maneuver the UGV and detect impediments in actual situations.**

Keywords*— *Camera/LiDAR Sensor Fusion, Obstacle Avoidance, Autonomous Navigation, Deep Learning, YOLOv7

I. Introduction

In recent years, sensor fusion methods have become increasingly significant, particularly in the context of autonomous vehicles and robotics. To detect obstacles with extreme accuracy in autonomous cars, it may be possible to integrate distance calculation with picture data from LiDAR and camera sensors. Therefore, such information might improve autonomous vehicle navigation [1, 2]. In this paper, we propose a novel camera and LiDAR fusion and autonomous navigation method for Unmanned Ground Vehicles (UGV).

Studies on sensor fusion design and object detection have been published in a number of journals. In [3], the detection points from 2-dimensional maps were projected onto the ground before using a method to group the objects. In [4], the segmentation was carried out using an unsupervised gaussian mixture model. In comparison to the 2-dimensional map technique, GMM uses direct LiDAR data processing, which is time-consuming but results in a greater separation rate between objects. However, segmentation outcomes depend on the predetermined number of clusters, making it difficult to achieve satisfactory results.

In this research, we suggest a fusion architecture for cameras and LiDAR sensors that weighs the advantages and disadvantages of the two sensors to create a detection system that is more dependable than others. In order to decrease the obstacle identification error rate and the problem of object occlusion, which frequently arises in camera-based object recognition, the suggested method makes use of a 3D point cloud received from the LiDAR depth sensors. The UGV's autonomous navigation system is then provided with the region proposal to re-plan its route and navigate appropriately. The experiments were conducted on a UGV system with high obstacle avoidance and fully autonomous navigation capabilities.

The results of the experiments show that the proposed method can effectively detect obstacles and navigate the UGV in real-world environments. The proposed approach has the potential to improve the performance and safety of autonomous vehicles and UGVs, by providing a reliable sensor fusion method for obstacle detection, distance estimation, and obstacle avoidance.

II. Methodology

This section goes into great length on the proposed camera and LiDAR fusion and autonomous navigation system for self-driving cars. The method records the different physical aspects of the environment using a camera and a LiDAR sensor in order to comprehend its surroundings. The algorithm's first phase entails calibrating the camera and LiDAR in order to estimate the devices' displacement, as seen in Figure 1. These actions are only carried out once or if a new calibration is required. The blue box visualizes this step. Then, YOLOv7 receives the monocular RGB images as input to find objects in the route of the vehicle. The fusion module is then supplied with the bounding box of identified objects and 3D LiDAR data points.

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The LiDAR point clouds are mapped onto the RGB images in this module based on the calibration parameters extracted in the previous module. In order to perform autonomous navigation and obstacle avoidance tasks, the SLAM module is given the distance of identified objects. Pixel-to-point matching of the sensor data and alignment score evaluation of the matched data are used to assess the accuracy and utility of the suggested approach.

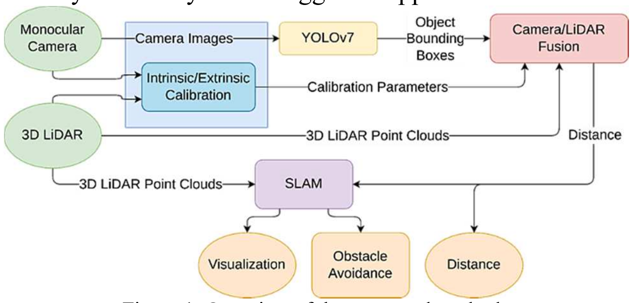


Figure 1. Overview of the presented method.

A. *Sensor Calibration*

Finding the intrinsic and extrinsic calibration parameters requires two steps of back-to-back calibration of the camera and LiDAR. Before obtaining the camera/LiDAR extrinsic parameters, the camera intrinsic parameters are first calculated using the checkerboard approach for camera calibration. Figure 2 and Figure 2.1. depicts the test vehicle's sensor configuration.

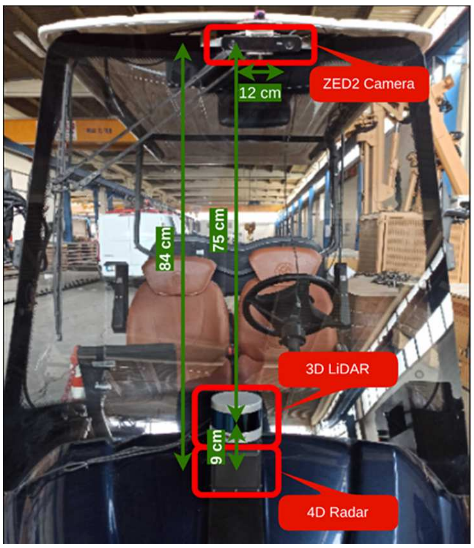


Figure 2. Sensor setup of the test vehicle: Front view.

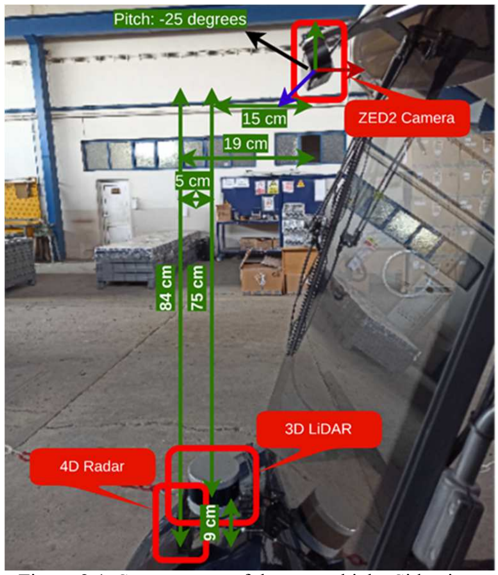


Figure 2.1. Sensor setup of the test vehicle: Side view.

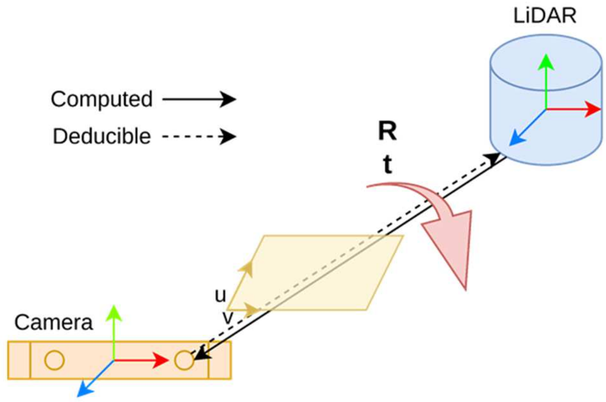


Figure 3. Problem statement of the extrinsic calibration of camera/LiDAR.

Figure 2 illustrates how sensors were fixedly attached to the front of the vehicle. This paper made use of a Microsoft ZED2 model camera and a Velodyne VLP-16 model LiDAR sensor. Figure 3 depicts the problem statement of extrinsic calibration of the camera/Lidar. Estimating the rotation R and translation t between two sensors' coordinate systems is referred to as computing the extrinsic calibration. We are able to determine the other direction by directly computing the extrinsic from the LiDAR to the camera. To do this, A 2D coordinate system (u, v) is used to represent the picture data from the camera, and a 3D coordinate system is used to display the 3D point cloud created from the raw measurements of the LiDAR and radar sensors (x, y, z). The computation of the projective transformation matrix, which projects the 3D LiDAR and radar points on the 2D picture, is the primary goal of the camera/LiDAR and camera/radar calibration. The following equation can be used to perform the transformation between 2D image pixels and 3D point cloud points:

$$I_i \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{bmatrix} f & 0 & u_0 & 0 \\ 0 & f & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} P_i \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} \quad (1)$$

where I_i is the image pixel (u, v), f is the focal length of the camera, point (u_0,v_0) is the center point of the image plane, (r_11,...,r_33) are the elements of the rotation matrix, and (t_x,t_y,t_z) is the translation vector. Finding the 4×4 transformation matrix between coordinate systems relates to the extrinsic calibration between camera/LiDAR. We adopted Levinson and Thrun's method [7] for determining the extrinsic calibration between camera/LiDAR.

B. YOLOv7 Object Detection

We used the most recent YOLO object detection model till the writing of this paper, YOLOv7, for object detection. To match the distribution of the ground truth boxes from our training set, we selected bounding boxes. It should be obvious that there are only a few sorts of vehicles that are typically on the road because we are primarily focusing on automobiles and pedestrians. Given the restrictions, there is, in particular, little variance in the width of these classes. A simple architecture of YOLO object detection models is shown in Figure 4.

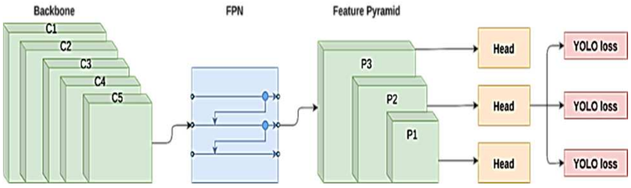
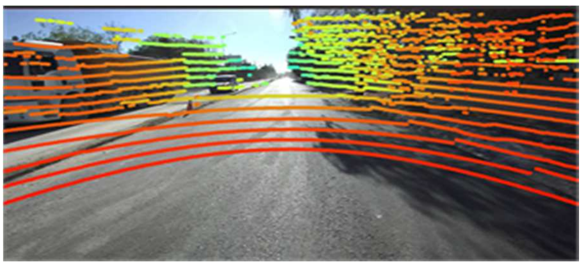


Figure 4. The simple architecture of the YOLO network.

C. Camera/LiDAR/Radar Fusion

After calculating the intrinsic and extrinsic parameters, Equation (1) is used to project the LiDAR data points onto the camera image. Figure 5 depicts a case of this fusion as an illustration. Building a thorough understanding of the autonomous vehicle requires the effective fusion of data obtained from various sensors. The redundant information from the sensors can be effectively used by the data fusion process to understand the surrounding environment. Despite the great efficacy and dependability of camera-based object recognition and classification, it might be difficult to pinpoint the exact depth of objects that have been identified. In contrast, the LiDAR sensor performs well at determining an object's depth but struggles to classify nearby and small objects. Using only a camera or LiDAR to gauge the object's depth is very challenging. Consequently, object detection and classification in three dimensions are greatly enhanced by the fusion of the LiDAR and camera outputs.



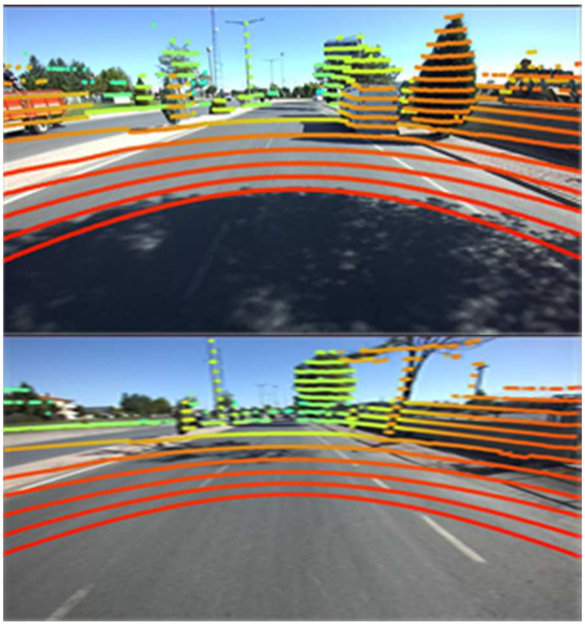


Figure 5. LiDAR data point alignment on RGB images.

The sensor fusion approach uses data from the camera and LiDAR sensor to calculate object distance. The overview of the estimation of the object distance is shown in Figure 6. The calibrated LiDAR 3D points aligned inside the bounding box as seen by the camera are used to extract the depth of the object based on the sensor fusion outcome. The method used for object identification in the camera is the same as that described in the publication [X].

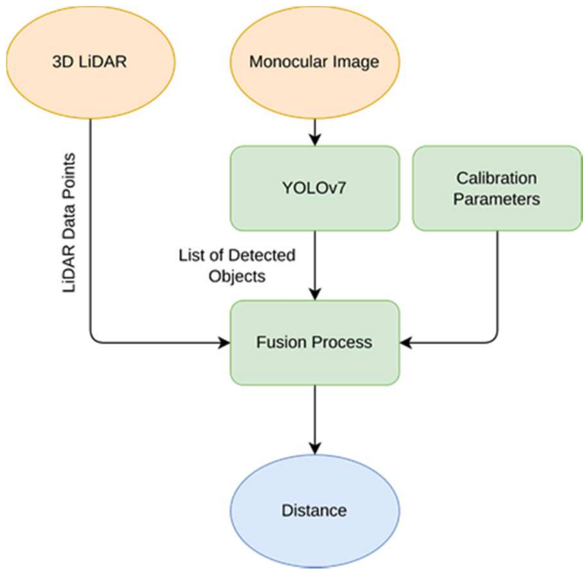


Figure 6. Overview of the fusion module of the system.

In Figure 6, the input to the fusion module is the data point from LiDAR, a list of detected objects in RGB image using YOLOv7, and the calibration parameters calculated in the previous step. The output data is the distance of detected objects. The detected objects are represented using bounding boxes. Equation 2 is used to extract projected LiDAR points inside the bounding boxes and estimate the distance of the detected objects.

$$p_i^{extracted} = \{p_i \in f_t^{2DLiDAR}(x, y, p) \\ if\ p_i\ inside\ bounding\ box(x_1, y_1, x_2, y_2)\} \tag{2}$$

where $p_i^{extracted}$ is the data points of LiDAR that are inside the bounding boxes of detected objects. The bounding boxes are represented by two points (x_1, y_1, x_2, y_2), the top left point and bottom right point of the rectangle. After extracting the points included inside the boundary boxes, they are used to calculate the distance of the objects. These data are calculated using the following equation where d_{obj} is the distance of the object.

$$d_{obj} = \{min\_z(center\_quartile(p_i^{extracted}(x, y, z) : i = 1, \dots, n))\} \tag{3}$$

III. EXPERIMENTAL RESULTS

For the purpose of estimating distance for a self-driving vehicle, we discuss the entire experimentation and findings of the sensor fusion method in this section. The sensor fusion approach was assessed by taking into account the various facets of environmental sensing in accordance with the present requirements of self-driving cars.

The self-driving vehicle and hardware configuration used in this study are depicted in Figure 7. A Microsoft ZED2 stereo camera for streaming video, a Velodyne VLP-16 3D LiDAR sensor, a SmartMicro UMRR-11 TYPE 132 radar sensor, and both for 3D acquisition, a 2D RPLiDAR for 2D acquisition, a BNO055 inertial measurement sensor, and an RTK-enabled FOIF GNSS were all part of the vehicle platform. We only used a camera and the 3D LiDAR in this study. It should be noted that, despite the fact that the ZED2 is a stereo camera, only one left monocular was used in this study. The LiDAR and camera sensors were installed in the front of the car, as shown in Figure 2. The camera and LiDAR sensors were calibrated to estimate the intrinsic and extrinsic parameters before sensor fusion was acquired, as detailed in Section 2.

Figure 7. The self-driving golf vehicle and its sensors used in this paper.

Figure 7.1. Sample shots from camera, LiDAR and radar sensors.

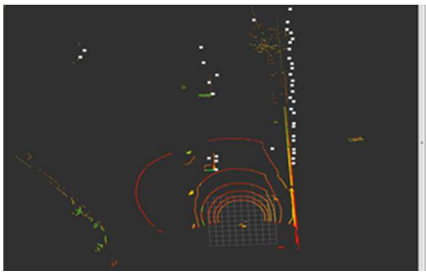


Figure 7.2. LiDAR and radar points are shown in color and white, respectively.

The experimental results of the sensor fusion technique for object distance estimation for autonomous automobiles are shown in Figure 8. In this test, data points that were aligned with RGB images were used to estimate the distances between items in the surrounding area. The points inside the bounding box of the identified objects were used to determine the objects' distance. The distance of the objects was estimated by considering the minimum distance of the center quartile of the detected objects' bounding boxes. The YOLOv7 object identification was out of the scope of the suggested approach, hence it is not covered in detail in this study. This experiment aims to assess the usefulness of the data fusion method in comparison to the single camera or LiDAR-based strategy for measuring the distance in self-driving cars because the usual vision-based distance estimate methods are not reliable enough. Despite the fact that LiDAR-based systems offer incredibly accurate distance estimations, they take a long time and don't have the same level of accuracy as vision-based object detections. This experiment shows numerous settings where the distance of objects can be estimated by combining the precise distance measurement provided by LiDAR with the recognition capability of the YOLOv7.

Figure 8. Experimental results of sensor fusion method (Camera/LiDAR).

Figure 8.2. Experimental results of sensor fusion method (Camera/LiDAR).

Figure 8.1. Experimental results of sensor fusion method (Camera/LiDAR).

Table 1 presents the numerical outcomes of a few scenes from the experimental tests in the on-road scenarios. Six arbitrary situations with vehicles and pedestrians at different distances were taken into account for the tests. Their corresponding snapshots are shown in Figure 8. The results of the table show that the suggested fusion technique was capable of estimating object distances even at distances of more than 50 m. Additionally, the occluded pedestrians and vehicles in scenes 1, 2, and 6 ensure the distance measurement capability of the method in the occluded scenarios too. In light of this, The sensor fusion method seems to be more accurate than using a camera or a LiDAR sensor alone, especially for hidden objects.

Scene	Objects	Distance
1	Vehicle 1	8.692
	Vehicle 2	18.98
2	Vehicle 1	16.60
	Vehicle 2	30.46
	Vehicle 3	39.31
	Vehicle 4	30.69
	Vehicle 5	25.07
3	Vehicle 1	15.66
	Vehicle 2	18.48
4	Vehicle 1	9.277
	Vehicle 2	17.06
5	Vehicle 1	27.85
6	Pedestrian 1	6.582
	Pedestrian 1	7.056
	Vehicle 1	43.01

Finally, we have tested the proposed sensor fusion method in obstacle avoidance and autonomous navigation of our test vehicle. In figure 9, the three sample sequential shots (left to right) from the test are shown. The top images show the camera view and the bottom images demonstrate the real-time SLAM information from sensors. As can be seen, the vehicle in the left image detected two people in front of the vehicle and replanned the route to the goal position. In the middle image, the vehicle is autonomously avoiding obstacles and navigating toward the goal position. In the right image, the vehicle has reached the goal position.

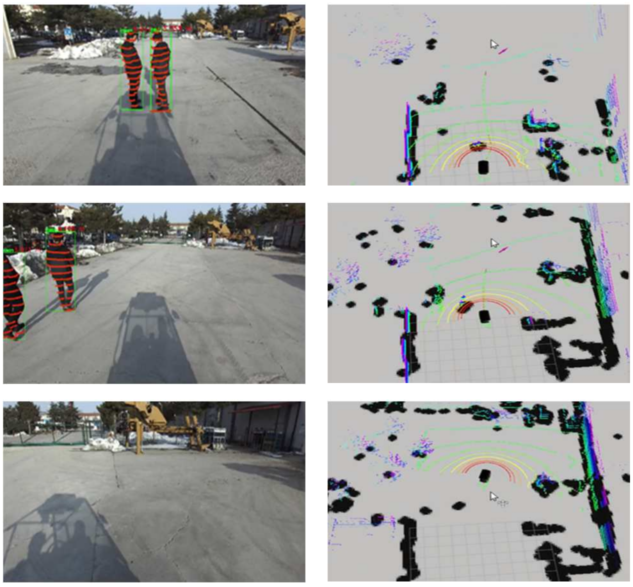


Figure 9. The 3 sample sequential (left to right) shots from the autonomous navigation test.

IV. CONCLUSIONS

In conclusion, this research proposed a novel camera and LiDAR fusion and autonomous navigation method for Unmanned Ground Vehicles (UGV). The proposed system balances the benefits and drawbacks of the two sensors to produce a detection system that is more reliable than others. By utilizing a 3D point cloud returned from the LiDAR depth sensors, the system is able to further examine the shape of objects and lower the obstacle detection error rate and reduce the issue of object occlusion. The experiments conducted on a UGV system with high obstacle avoidance and fully autonomous navigation capabilities have shown that the proposed method can effectively detect obstacles and navigate the UGV in real-world environments. The findings of this research might raise the standard of performance. and safety of autonomous vehicles and UGVs, by providing a reliable sensor fusion method for obstacle detection, distance estimation, and obstacle avoidance.

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REFERENCES

[1] Gu, Shuo, et al. "Road detection through crf based lidar-camera fusion." 2019 International Conference on Robotics and Automation (ICRA). IEEE, 2019.

[2] El-Sheimy, Naser, and You Li. "Indoor navigation: State of the art and future trends." Satellite Navigation 2.1 (2021): 1-23.

[3] Himmelsbach, Michael, Thorsten Luettel, and H-J. Wuensche. "Real-time object classification in 3D point clouds using point feature histograms." 2009 IEEE/RSJ International Conference on Intelligent Robots and Systems. IEEE, 2009.

[4] J Maligo, Artur, and Simon Lacroix. "Classification of outdoor 3D lidar data based on unsupervised Gaussian mixture models." IEEE Transactions on Automation Science and Engineering 14.1 (2016): 5-16.

[5] Yin, Huilin, Xiaohan Yang, and Chao He. "Spherical coordinates based methods of ground extraction and objects segmentation using 3-D LiDAR sensor." IEEE Intelligent Transportation Systems Magazine 8.1 (2016): 61-68.

[6] Kusenbach, Michael, Michael Himmelsbach, and Hans-Joachim Wuensche. "A new geometric 3D LiDAR feature for model creation and classification of moving objects." 2016 IEEE Intelligent Vehicles Symposium (IV). IEEE, 2016.

[7] Levinson, Jesse, and Sebastian Thrun. "Automatic online calibration of cameras and lasers." Robotics: science and systems. Vol. 2. No. 7. 2013.